

Project ASTRA

Product Requirements Document (PRD)

Subtitle: Agentic AI Investigation Platform for Karnataka State Police

Version: 1.1 (Hackathon MVP - Architecture Revised)

1. Executive Summary

Project ASTRA is an AI-powered Investigation Copilot built for the Karnataka State Police (KSP) to accelerate criminal investigations using conversational AI, retrieval-augmented generation (RAG), and intelligent relationship analysis.

Instead of requiring officers to manually search multiple systems, ASTRA enables investigators to ask natural language questions such as:

"Show all cyber fraud cases linked to this UPI ID."

The platform automatically retrieves relevant FIRs, discovers relationships between suspects, vehicles, phones, bank accounts, and evidence, generates investigation summaries, and recommends next investigative actions.

The MVP is designed specifically for the KSP Datathon and is built natively on Zoho Catalyst, leveraging Catalyst's serverless architecture, AI services, secure storage, and orchestration capabilities.

2. Problem Statement

Investigating officers currently spend significant time navigating multiple databases, manually correlating evidence, and generating reports.

Current challenges include:

- Slow FIR retrieval
- Manual cross-case relationship analysis
- Fragmented investigation workflows
- Time-consuming report generation
- Limited conversational search capabilities
- Poor visibility into cross-jurisdiction crime patterns

These challenges increase investigation time and reduce operational efficiency.

3. Product Vision

Build an AI-first investigation platform where every police officer can interact with crime data using natural language.

The AI should function as an intelligent investigation assistant capable of reasoning over structured and unstructured police data, orchestrating multiple investigative tasks, and providing explainable recommendations.

4. Goals

Primary Goals

- Reduce investigation latency
- Simplify access to crime intelligence
- Improve cross-case relationship discovery
- Reduce manual report generation
- Demonstrate production-ready AI architecture on Zoho Catalyst

Success Metrics

Metric	Target
AI Voice Pipeline Response Time	8-15 seconds
Text Query Perceived Latency	< 1 second (via Server-Sent Events / SSE)
FIR Retrieval	< 2 seconds
Relationship Analysis	< 5 seconds
Summary Generation	< 10 seconds
Retrieval Accuracy	> 90%
Supported Dataset	100K+ synthetic FIRs

5. Target Users

Investigating Officer

- Search FIRs
- Retrieve evidence
- Generate summaries
- Ask investigative questions

Police Inspector

- Monitor station cases
- Review investigations
- Track officer workload

Superintendent of Police (SP)

- District-level analytics
- Crime trends
- Resource allocation

Senior Officers (DIG / IGP / DGP)

- Statewide dashboards
- Crime intelligence
- Strategic insights

6. MVP Scope

Included

- Conversational AI Assistant
- Natural Language Crime Search
- FIR Summarization
- Relationship Graph
- Cyber Fraud Intelligence
- Crime Analytics Dashboard
- Voice Interaction (English & Kannada)
- Evidence Metadata Search
- Investigation Report Generation

Out of Scope

- Live CCTNS integration
- Citizen-facing portal
- Court management
- Biometric authentication
- Production police databases

7. Functional Requirements

FR-001 — Conversational AI

The system shall allow officers to interact using natural language.

Capabilities:

- Multi-turn conversations
- Context retention
- Follow-up questions
- English support
- Kannada support
- Voice input
- Voice output
- Client-side PDF Session Export (via jspdf bypassing server rendering overhead) to satisfy local export mandates

FR-002 — Crime Search

Search using:

- FIR Number
- Person Name
- Phone Number
- Aadhaar (synthetic)
- Vehicle Number
- Bank Account
- UPI ID
- Address
- Crime Type
- Police Station
- District

FR-003 — FIR Summarization

Generate structured summaries including:

- Incident overview
- Timeline
- Victims
- Accused
- Evidence
- Witnesses
- Investigation status
- Recommended next actions

FR-004 — Relationship Intelligence

Automatically discover links between:

- Persons
- FIRs
- Vehicles
- Phones
- Bank Accounts
- UPI IDs
- Addresses
- Organizations

Display results as an interactive graph.

FR-005 — Crime Analytics

Dashboard widgets include:

- Crime by district
- Crime trend over time
- Cyber fraud hotspots
- Officer workload
- Investigation progress
- Repeat offenders
- Common Modus Operandi
- Socio-Economic & Demographic Correlation (mapping economic stress/urban density against crime rates)

FR-006 — Evidence Search

Search evidence by:

- Case ID
- Image metadata
- Audio metadata
- Video metadata
- Officer
- Date

FR-007 — AI Investigation Report

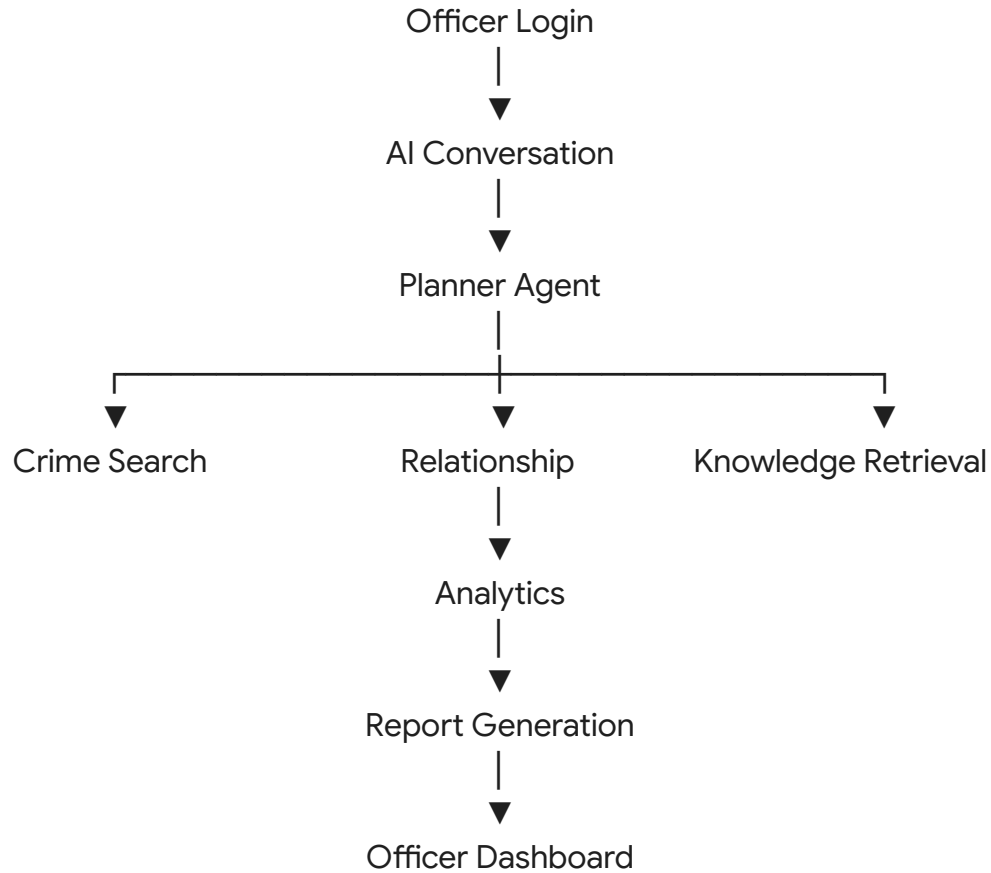
Generate downloadable reports containing:

- Investigation summary
- Related cases
- Similar crimes
- Timeline
- Recommendations

FR-008 — Explainable AI Lineage

All LLM response payloads must enforce a schema containing a `source_nodes` array with specific `fir_id` or `evidence_id` foreign keys, rendered as deterministic clickable reference tags.

8. User Journey



9. AI Multi-Agent Architecture

Conversation Agent

- Understands user intent
- Maintains conversation context
- Handles follow-up questions

Planner Agent

Breaks a complex request into subtasks.

Example:

"Find cyber fraud linked to this account."

- Search FIRs
- Search bank accounts

- Search UPI IDs
- Retrieve evidence
- Build relationship graph
- Generate report

Crime Search Agent

Queries structured investigation data.

Knowledge Agent

Retrieves relevant SOPs, FIR context, and legal references using RAG.

Relationship Agent

Builds a knowledge graph connecting people, accounts, devices, vehicles, and FIRs.

Analytics Agent

Generates:

- Crime trends
- Heatmaps
- Repeat offender analysis
- Modus Operandi insights

Report Agent

Produces structured investigation reports.

10 .Zoho Catalyst Architecture

Catalyst Authentication

Responsible for:

- Officer Login
- JWT Authentication
- Session Management
- Role-Based Access Control

Catalyst AppSail

Hosts the backend API.

Responsibilities:

- REST APIs
- Agent orchestration

- Dashboard APIs
- Conversation management
- Implement Server-Sent Events (SSE) from the Node.js backend to stream token generation incrementally to the client.

Advanced I/O Functions

Used for low-latency serverless operations.

Examples:

- Search FIR
- Search Person
- Search Vehicle
- Retrieve metadata
- Validate requests

Job Functions

Handles long-running AI tasks.

Must handle deep graph traversals. Architecture dictates an asynchronous polling pattern: Initial client request returns HTTP 202 Accepted with a unique `job_id`.

Frontend state manager executes exponential backoff polling against a lightweight `GET /api/v1/jobs/{job_id}/status` endpoint while the 15-minute Job Function writes computed graph JSON to Stratus object storage.

Examples:

- Report generation
- Relationship analysis
- Batch summarization
- Analytics computation

Circuits

Coordinates multi-step AI workflows.

Example workflow:

Conversation → Planner → Search → RAG → Relationship Analysis → Report-Generation → Response

QuickML LLM Serving

Provides managed LLM inference.

Responsibilities:

- Reasoning
- Conversation
- Summaries
- Investigation recommendations

QuickML RAG

Retrieves contextual information from indexed documents.

Knowledge sources:

- FIRs
- Investigation manuals
- BNSS
- SOPs
- Historical crime data

Catalyst Data Store

Stores structured data.

Core Tables:

- Users
- FIRs
- Persons
- Victims
- Accused
- Vehicles
- Phones
- Bank Accounts
- UPI IDs
- Evidence Metadata
- Conversations

ZCQL

Used for querying structured crime data efficiently.

Object Storage (Stratus)

Stores unstructured evidence.

Examples:

- Images
- Videos

- PDFs
- Audio
- Investigation documents

Connections

Secure integration layer for future APIs such as:

- CCTNS
- NCRP
- I4C
- Internal police systems

Logs & Metrics

Used for:

- Monitoring
- Debugging
- Performance tracking
- Audit trails

11. Zoho Service Mapping

Feature	Zoho Catalyst Service
Officer Login	Catalyst Authentication
Frontend Hosting	AppSail
REST APIs	AppSail
Business Logic	Advanced I/O Functions
Agent Workflow	Circuits
Long-running Tasks	Job Functions
AI Reasoning	QuickML LLM Serving
Knowledge Retrieval	QuickML RAG
Crime Database	Data Store
Structured Queries	ZCQL
Evidence Storage	Object Storage (Stratus)
External APIs	Connections
Logs	Catalyst Logs
Monitoring	Catalyst Metrics

12. Database Overview

- Users
- PoliceStations
- FIRs
- Persons / Accused: Requires demographic columns (employment_status, income_bracket, urban_density_index) for Analytics Agent aggregations.
- Victims
- Vehicles
- Phones
- BankAccounts
- UPIs
- Evidence
- Organizations
- Conversations
- Edges (Adjacency List): Maps source_node_id to target_node_id to optimize multi-degree relationship hops and bypass nested ZCQL relational join bottlenecks.

13. API Endpoints

Method	Endpoint	Description
POST	/api/v1/conversations	AI Conversation
POST	/voice	Voice Query
GET	/api/v1/firs	Search Crime Database (with query parameters for searching)
GET	/fir/{id}	Retrieve FIR
GET	/api/v1/entities/{id}/network	Relationship Graph
GET	/analytics	Dashboard Data
POST	/api/v1/firs/{id}/summarize	Generate Investigation Summary

GET	/dashboard	Officer Dashboard
GET	/api/v1/jobs/{job_id}/status	Async Job Status Polling

14. Security

- JWT Authentication
- Role-Based Access Control
- Audit Logging
- Encrypted Object Storage
- Secure API Gateway
- Least Privilege Access

15. Demo Workflow

Scenario:

An officer receives a complaint regarding a suspicious UPI transaction.

The officer asks:

"Find all cyber fraud cases linked to UPI ID XXXX."

ASTRA performs the following:

1. Authenticates the officer.
2. Searches the Data Store for matching FIRs.
3. Retrieves related evidence from Object Storage.
4. Uses QuickML RAG to retrieve similar historical cases.
5. Builds a relationship graph connecting suspects, bank accounts, phones, and FIRs.
6. Generates an investigation summary.
7. Displays related cases, hotspots, and recommended next investigative actions.

16. Future Enhancements

- Live CCTNS integration
- ICJS integration
- Predictive crime analytics
- Image similarity search
- Facial recognition integration
- Automatic FIR drafting

- Mobile application
- Real-time investigation alerts

17. Acceptance Criteria

The MVP will be considered successful if it can:

- Authenticate officers securely.
- Answer natural language investigative queries.
- Retrieve crime records accurately.
- Generate FIR summaries.
- Discover relationships between entities.
- Display investigation dashboards.
- Generate downloadable reports.
- Demonstrate an end-to-end investigation workflow using Zoho Catalyst services.

18. Conclusion

Project ASTRA demonstrates how an agentic AI platform built entirely on Zoho Catalyst can transform police investigations. By combining conversational AI, retrieval-augmented generation, relationship intelligence, and secure cloud-native architecture, the platform reduces investigation effort while improving access to actionable intelligence.

The MVP is intentionally scoped for the KSP Datathon while maintaining a production-ready architecture that can scale to statewide deployment.